SOMA: Strategic Orchestration and Memory-Augmented Agentic System for Zero-Shot VLA Generalization

Abstract— **Despite the promise of Vision-Language-Action (VLA) models as generalist robotic controllers, their zero-shot generalization to out-of-distribution (OOD) tasks is fundamentally hindered by an "amnesic" limitation, where execution traces are treated as isolated events, discarding critical historical experiences. To address this, we propose SOMA, a Strategic Orchestration and Memory-Augmented system that transforms standalone VLAs into cognitively-aware agents via a self-evolving 'Brain-Eyes-Limbs' recursive architecture. Specifically, the Eyes provide contextual awareness via experience retrieval, the Brain dictates strategic interventions through failure attribution, and the Limbs implement the Brain-generated, dynamic MCP tool chains. The loop is closed as new experiences are committed to the memory bank, where a two-stage iterative attribution process refines the stored knowledge for long-term performance enhancement. Experimental evaluations across diverse base models demonstrate that SOMA significantly enhances zero-shot resilience in OOD environments without parameter fine-tuning, providing a scalable pathway toward adaptive robotic foundation models. Project page and source code are available at: `https://anonymous.4open.science/r/SOMA-122712`.**

I. INTRODUCTION

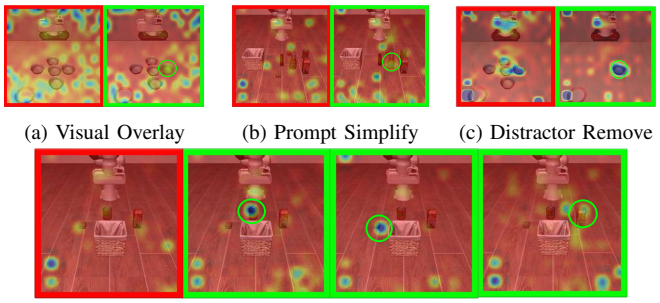
Vision-Language-Action (VLA) models have revolutionized end-to-end robotic manipulation by bridging multimodal perception, linguistic grounding, and continuous control within a unified policy. In practical applications, these models function much like a robot's 'cerebellum'—efficiently instantiating high-level intents into sensorimotor actions [1], [2]. However, as service and industrial robots transition from structured laboratories to the unpredictability of the open world, the primary challenge has shifted: the goal is no longer just precise in-distribution execution, but robust zero-shot generalization [3]. While traditional fine-tuning remains effective for specialized scenarios, the capacity for zero-shot generalization is emerging as a critical bottleneck for general-purpose robots, as real-world deployment necessitates handling infinite corner cases that cannot be exhaustively pre-collected.

This bottleneck is primarily rooted in the structural inability of static VLA policies to accommodate the inevitable out-of-distribution (OOD) conditions [4], [5], [6] encountered in the real world. In these settings, OOD shifts arise from novel object appearances, shapes, and materials [7], [8]; changes in illumination, camera viewpoints, and sensor configurations; and linguistic variability and ambiguity in user instructions [9], [10], [11]. These factors jointly create a setting in which the data distribution is effectively unbounded, the environment is partially observable and non-stationary, and failures are often caused by coupled errors across perception, language grounding, and control.

To address these challenges, prior work has progressively expanded the generalization capacity of VLA systems along multiple tightly coupled axes. Early advances focused on large-scale vision-language pretraining followed by lightweight fine-tuning on limited robot trajectories, enabling transferable representations that support cross-task and cross-object generalization, as exemplified by RT-2 [4] and OpenVLA [12]. Building on this foundation, subsequent efforts emphasized robust multimodal representation learning, leveraging invariant visual encoders (e.g., SigLIP [13], DINOv2 [14]), large language models (e.g., T5 [15], LLaMA [16], [17]), and cross-modal alignment mechanisms such as contrastive learning and flow matching to mitigate distribution shifts induced by appearance variation and linguistic diversity. Recognizing the fundamental limitations imposed by narrow and biased robotic datasets, researchers further introduced data-centric generalization strategies, including multi-embodiment dataset aggregation (e.g., Open-X Embodiment [18]), synthetic augmentation across appearance, lighting, and backgrounds, and the exploitation of unlabeled human video (e.g., LAPA [19]) to extract latent action structures and expand behavioral coverage. More recently, advances in control and action generation have introduced diffusion- or flow-matching-based continuous action modeling (e.g., Octo [20], π_0 [21]), reducing reliance on fixed motion primitives and improving adaptability to unfamiliar action patterns. At a higher level of abstraction, hierarchical control architectures (e.g., RT-H [22], $\pi_{0.5}$ [23]) further decouple high-level task planning from low-level motor execution, enabling more flexible adaptation to task logic and environmental variation.

Despite these advances, a common implicit assumption remains: sufficiently capable models can "learn away" generalization failures. However, in open-world settings, system failures rarely arise from isolated perceptual errors; instead, they emerge from complex couplings among task misinterpretation, perceptual uncertainty, execution errors, and environmental feedback. Addressing such failures requires systems to possess *(i)* long-term memory for storing and reusing past experience, *(ii)* high-level causal attribution and strategic adjustment capabilities, and *(iii)* dynamic orchestration and coordination of external tools to actively intervene in the perception-action stream.

Current VLA models inherently lack these system-level capabilities, and thus, even with strong pretraining and multimodal alignment, their zero-shot generalization performance remains structurally bounded. Admittedly, certain funda-



(a) Visual Overlay (b) Prompt Simplify (c) Distractor Remove

(d) Long-Horizon

Fig. 1: **PaliGemma attention maps under different inputs.** Red and blue signify high and low attention, respectively. Baseline maps (red frames) display diffuse, non-specific attention. In contrast, targeted MCP tool interventions (green frames) facilitate precise target object identification, characterized by refined object-edge delineation indicated by localized blue attention prior to manipulation. This refined focus is achieved through: (a) overlaying visual cues; (b) simplifying linguistic prompts; (c) filtering background clutter; and (d) decomposing long-horizon subtasks.

mental OOD challenges may still necessitate specialized hardware-level redesigns or extensive post-training.

Nevertheless, we argue that many failures do not stem from a deficiency in basic motor skills. Our analysis of the attention maps in Fig. 1 reveals a critical paradox: while baseline models often exhibit diffuse and non-specific attention (red frames) in OOD settings, they still possess the latent capacity for precision. By applying targeted MCP tool interventions, we observe a striking transition to refined object-edge delineation (green frames).

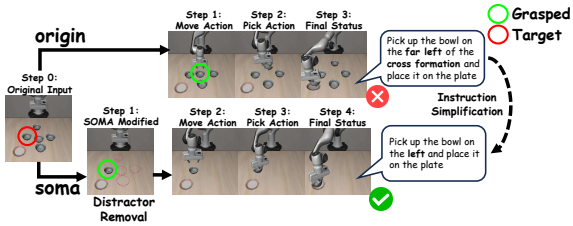
This visual evidence confirms that the model's failure is primarily a "perceptual drift" rather than an execution collapse: the frozen VLA "knows how" to move, but its focus is obscured by environmental or prompt-level noise. This insight suggests a significant, untapped potential within frozen VLAs that can be unlocked without costly retraining. This compels us to rethink the standard execution pipeline: by harnessing experiential knowledge to dynamically orchestrate these input-level interventions, we can actively re-align the agent's focus, thereby bridging the gap between historical mastery and novel challenges. Consequently, this leads to our primary research inquiry:

> *Can a systemic, agentic architecture transcend the inherent boundaries of static policies by orchestrating a cognitive loop that unlocks the latent potential of frozen VLAs?*

To address these limitations, we propose **SOMA**, a **S**trategic **O**rchestration and **M**emory-**A**ugmented agentic framework that upgrades a frozen VLA policy into a cognitively adaptive agent without parameter fine-tuning. Rather than relying solely on stronger representation learning, SOMA equips the policy with system-level capabilities for experience reuse, failure-aware reasoning, and dynamic intervention. Specifically, SOMA maintains a structured experience memory and retrieves the most relevant successes and failures for each context. A strategic LLM then performs theoretical attribution to infer root causes and dynamically composes *Model Context Protocol (MCP)* [24] tool chains, which, together with the base VLA, implement targeted interventions. The resulting trajectories and attributions are written back to memory, where the LLM further refines and consolidates experience, closing the loop and preventing repeated failure modes. We evaluate SOMA across multiple backbone VLA models and a suite of out-of-distribution generalization challenges. SOMA consistently improves zero-shot success rates over frozen backbones, demonstrating strong plug-and-play compatibility and robust gains without retraining. Through the diverse manipulation tasks illustrated in Fig. 2, we demonstrate that integrating SOMA with the π_0 backbone significantly boosts success rates under various OOD perturbations. Extending beyond this individual case, comprehensive evaluations across multiple robotic foundation models confirm that SOMA consistently enhances zero-shot resilience. Notably, in tasks involving fine-grained visual grounding and noisy linguistic prompts—where base models typically falter (0.0%–17.1% success rates)—SOMA elevates performance to a robust 74.9%–98.3% without parameter fine-tuning. Across all benchmarks, the framework achieves a remarkable average absolute success rate gain of over 68%, effectively quadrupling the baseline performance. Furthermore, SOMA maintains high temporal consistency in long-horizon task chaining, sustaining success rates above 89% where baseline performance degrades due to compounding errors.

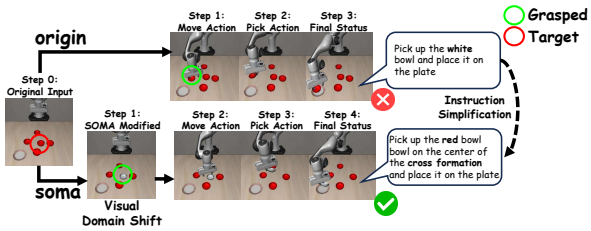
In summary, this work makes four contributions:

1) **System-Level Bottleneck Identification:** We identify a critical bottleneck in zero-shot VLA generalization: the gap between low-level control and high-level strategic reasoning. We show that monolithic policies underperform in certain OOD scenarios not because of inherent control boundaries, but due to an inability to effectively reuse historical experiences, resulting in a fragmentation of cognitive capabilities.
2) **A Plug-and-Play Agentic Framework:** We introduce SOMA, a modular, plug-and-play framework that decouples task complexity from the VLA policy. By leveraging the Model Context Protocol (MCP), SOMA allows for the seamless integration and expansion of tool chains, enabling the system to adapt to novel tasks without retraining the core controller.
3) **Recursive Self-Evolving Cognitive Cycle:** We propose a tripartite cycle of **retrieval, reasoning-execution, and consolidation**. Unlike static RAG systems, SOMA performs continuous self-iteration; it employs an LLM to perform post-hoc attribution on trajectories, recursively distilling raw experiences into high-level insights that optimize the long-term experience library.



(a) **Clutter Removal: Distractor Removal and Prompt Refinement.** SOMA simplifies the task by masking visual distractors and refining the instruction (from "far left of the cross formation" to "on the left") based on retrieved experiences, significantly enhancing target identification and grasping success.

(b) **Noisy Prompt: Instruction Refinement.** SOMA filters conversational noise from the redundant prompt ("Hey, umm…look down there…") and rewrites it into a concise command ("Pick up the red sauce bottle…") based on retrieved experiences. This linguistic simplification resolves ambiguity, ensuring precise target identification and success.



(c) **Visual Focus: Paint-to-Action and Prompt Refinement.** Faced with a complex "cross formation", SOMA re-renders the target red bowl into a white one and simplifies the instruction from "Pick up the red bowl…on the center" to "Pick up the white bowl". This dual-modality intervention resolves the baseline VLA's attention drift, enabling successful task execution in OOD scenarios.

Fig. 2: SOMA-Augmented Manipulation: Case Studies

4) **Zero-Shot OOD Generalization:** We demonstrate through evaluations across diverse tasks and backbone models that SOMA significantly enhances zero-shot generalization. By offloading reasoning to an evolving knowledge base and extensible tools, SOMA maintains robustness under diverse out-of-distribution conditions where traditional VLAs typically fail.

II. Related Work

In this section, we review recent advancements in Vision-Language-Action (VLA) models, retrieval-based methods in robotics, and agentic frameworks. We examine these approaches from a system-level perspective to identify potential bottlenecks in zero-shot OOD generalization, providing the context for the design of the SOMA architecture.

A. Vision-Language-Action Models and Generalization Challenges

Recent years have witnessed a paradigm shift in robot learning, moving from specialized policy networks to general-purpose Vision-Language-Action (VLA) models. Pioneering works such as π_0 [21] and OpenVLA [12] utilize massive web-scale datasets (e.g., Open-X Embodiment [18]) to enable end-to-end robotic control. By integrating robust visual encoders (e.g., SigLIP [13]) with large language models, these foundation models act as a powerful "cerebellum", capable of executing diverse instructions within their training distribution.

However, a critical limitation remains: these models operate as stateless, feed-forward networks that exhibit significant fragility when facing Out-of-Distribution (OOD) scenarios. To address this, approaches like X-VLA [25] attempt to adapt to new tasks via fine-tuning mechanisms (e.g., soft prompts). While effective, such methods are computationally prohibitive—for instance, pre-training X-VLA-0.9B requires approximately 64 NVIDIA A100 GPUs over 4 days [25]—and lack real-time adaptability, as they necessitate retraining for every novel concept not covered by the soft prompt. This highlights the necessity for a parameter-efficient, plug-and-play approach that enhances generalization without weight updates.

B. Retrieval-Augmented and In-Context Learning in Robotics

To mitigate data scarcity and improve adaptability without fine-tuning, Retrieval-Augmented Generation (RAG) and In-Context Learning (ICL) have been introduced to the robotics domain. Frameworks like RICL [26] and MemER [27] utilize vector databases to retrieve relevant historical demonstrations, using them to guide policy execution.

While these methods validate the utility of external memory, they suffer from a fundamental *Success-Bias*. Approaches like RICL predominantly retrieve *successful* action trajectories for passive imitation (interpolation). They overlook the high diagnostic value of *failure cases*. A potential limitation in existing retrieval-augmented robotics frameworks is the primary focus on *Action-Level Imitation*, which emphasizes trajectory replication (the "how") but may offer less transparency regarding the underlying decision logic (the "why"). For example, a system conditioned on a "red cup" trajectory might encounter challenges when the target attribute changes to "blue" if the retrieval process does not account for specific visual variations. To address this, SOMA incorporates a memory structure that considers both successful and unsuccessful experiences. By utilizing a Large Language Model (LLM) to categorize these precedents—such as distinguishing between perceptual mismatches and planning deviations—the framework aims to provide more context-aware guidance for the Vision-Language-Action (VLA) policy.

C. Zero-Shot Agentic Frameworks and Tool Use

Beyond passive retrieval, recent research has explored leveraging LLMs as high-level planners that utilize external tools. Pioneering works like Code-as-Policies [28] generate executable Python code for hierarchical tasks, while VoxPoser [29] grounds language instructions into 3D value maps to guide motion planners. Similarly, MOKA [30] utilizes Visual Prompting to annotate affordances directly on images.

Despite their efficacy in task planning, these frameworks encounter certain challenges in closed-loop adaptation:

- **Experience Utilization:** Systems such as VoxPoser [29] often operate as *zero-start* reasoners for each episode. Without a long-term memory mechanism to recall historical failure modes, such models may encounter difficulties in correcting repetitive errors across sequential trials.
- **Perceptual Constraints:** Many existing planners treat the perception module as a static input. In scenarios involving domain shifts (e.g., color bias), the planner may struggle to resolve underlying visual misinterpretations without active perceptual adjustments.
- **Tool Integration:** Tool invocation in methods like MOKA [30] is frequently based on predefined rules, which may limit the flexibility needed to dynamically select interventions based on real-time failure diagnosis.

To bridge these gaps, we propose SOMA, an integrated cognitive architecture that transcends simple RAG-based planning. By leveraging a dual-memory retrieval (successes and failures) for contrastive causal attribution, SOMA enables the system to pinpoint and rectify execution errors beyond zero-start reasoning. This process is bolstered by a two-phase evolution strategy (online adaptation and offline refinement) for long-term capability growth. Finally, SOMA utilizes the Model Context Protocol (MCP) to dynamically orchestrate extensible, composable intervention chains, validated via blank inference to ensure robust, adaptive closed-loop manipulation.

III. Methodology

In this section, we present the design of the **SOMA framework**, an agentic system that enables adaptive task execution for frozen Vision-Language-Action (VLA) models. As illustrated in Fig. 3, SOMA is structured around a **Tripartite Core** (Eyes, Brain, and Limbs) and a self-improving **Evolution** mechanism. Specifically, the Trinity Architecture structures the agent's real-time behavior:

1) **Eyes (Dual-Memory RAG)**: Retrieves historical success guidance and failure warnings to provide task-relevant context, acting as a visual-semantic reference frame that provides directional guidance for the LLM's decision-making.
2) **Brain (Attribution-Driven LLM Orchestrator)**: Diagnoses execution obstacles and orchestrates targeted interventions.
3) **Limbs (MCP Intervention for Agentic Execution)**: An extensible toolset that executes orchestrated strategies to modulate perceptual and linguistic inputs, directly facilitating the VLA's reasoning and execution.

Synthesizing these components, the **Evolution** mechanism (**Dual-Stage Memory Consolidation**) governs the long-term intelligence of the system. It bridges the **Brain's** reasoning with the **Eyes'** memory by continuously distilling new experiences into the dual-memory bank. Together, these elements constitute a self-reinforcing closed-loop process:

$$\textit{Observe} \rightarrow \textit{Retrieve} \rightarrow \textit{Diagnose} \rightarrow \textit{Orchestrate} \rightarrow \textit{Execute} \rightarrow \textbf{\textit{Consolidate}},$$

It enables real-time adjustments and long-term learning across different tasks. This biological analogy encapsulates the essence of **SOMA**: precision-driven contextual anchoring and directional steering through dual-memory retrieval (**Eyes**); strategic orchestration via attribution-driven inference and dynamic MCP scheduling (**Brain**); extensible intervention capabilities (**Limbs**); and continuous systemic evolution within a modular, plug-and-play architecture. We next describe each component in detail.

A. The Eyes: Dual-Memory RAG

The **Eyes** module implements a retrieval-augmented memory bank that provides historical context for downstream reasoning, functioning as a visionary reference to anchor the LLM's decision-making with past experiences. Unlike conventional RAG systems that retain only successful trajectories, our memory design is **Dual** along three complementary dimensions: (1) it stores both successful and failed experiences to enable contrastive attribution; (2) it adopts a semi-schema representation that combines fixed core fields with extensible metadata; and (3) each experience undergoes two stages of attribution, a process detailed in section III-D, which formally defines the Evolution mechanism.

We first describe the construction of the dual-memory experience bank, and then present the indexing and retrieval mechanism built upon it.

1) Dual-Memory Experience Bank Construction: We construct a dual-partition memory bank:

$$\mathcal{M} = \mathcal{M}_{succ} \cup \mathcal{M}_{fail} \tag{1}$$

which archives $\mathcal{M}_{succ}$ as positive guidance and $\mathcal{M}_{fail}$ (with diagnostic insights) as negative evidence. This duality allows the agent to reason by contrasting optimal paths with patterns to be avoided.

2) Semi-Schema Experience Representation: Each experience entry $e_i \in \mathcal{M}$ is stored in a **semi-schema format**, consisting of a fixed core schema augmented with extensible auxiliary fields:

$$e_i = \big(\overbrace{E_i, T_i, S_i, D_i, V_i, O_i, A_i}^{\text{fixed schema}}, \mathcal{I}_i\big) \tag{2}$$

- E_i: multi-modal embedding of the task instance;
- T_i: natural language task description;
- $S_i \in \{0, 1\}$: success indicator;
- D_i: diagnostic annotation (non-empty for failures);
- V_i: keyframe representations of the execution video;

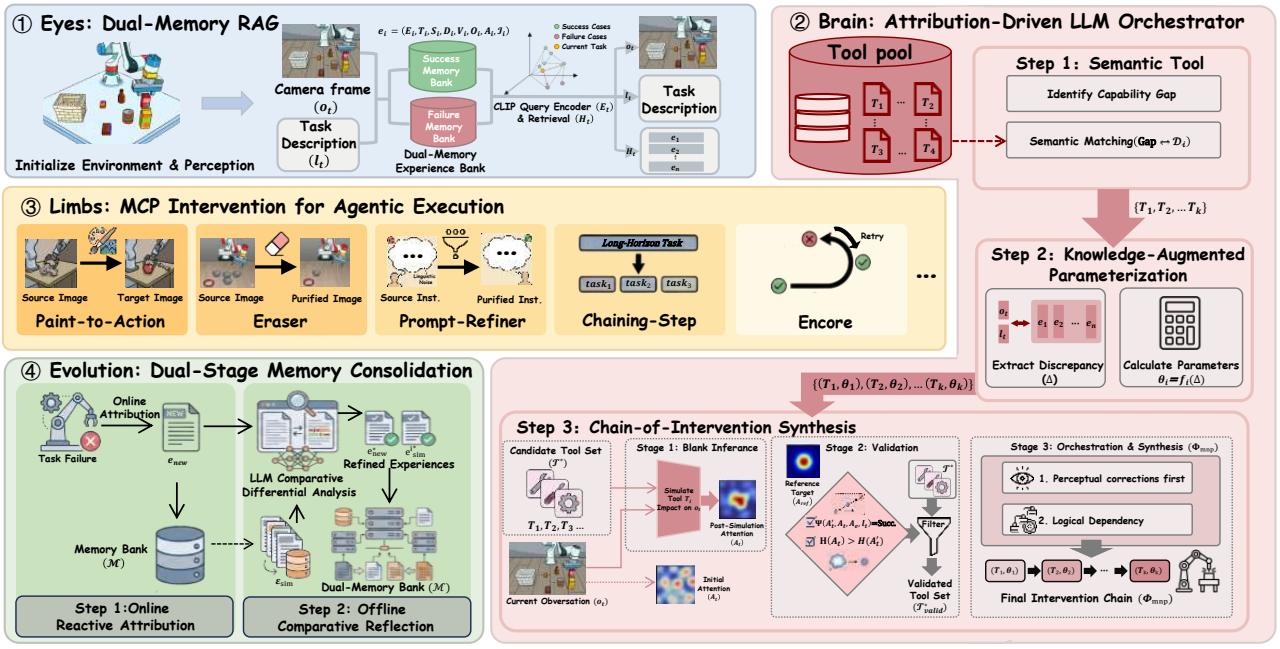


Fig. 3: **SOMA Framework:** SOMA is a pluggable, zero-shot reasoning framework designed for adaptive task execution in embodied agents. Given the current observation and instruction, the **Eyes** module first retrieves similar historical success or failure experiences from a dual-memory experience bank. Based on the retrieved context, the **Brain** module performs strategic orchestration by semantically matching the MCP toolset, calculating execution parameters θ, and synthesizing an optimal intervention chain through entropy-based validation and logical sequencing. These orchestrated instructions are then executed by the extensible **Limbs** module. Upon task completion, the **Evolution** module conducts dual-stage memory consolidation—comprising online reactive attribution and offline comparative reflection—to store the new experience back into the bank, ultimately forming a continuously iterating and evolving cognitive closed-loop.

- O_i: structured scene as object-attribute pairs.
- A_i: attention map: spatial distribution of VLA focus.
- $\mathcal{I}_i$: extensible structured metadata bundle.

A_i enable the Brain to diagnose visual biases by quantifying heatmap displacement and entropy shifts between the current OOD scene and retrieved references. This diagnostic precision facilitates targeted MCP interventions.

Complementing this, the metadata bundle $\mathcal{I}_i$ provides a flexible schema for tool execution, containing parameters such as `key_frame_range`, `success_max_step`, and `rollback`. Additional fields can be dynamically appended as new tools or reasoning signals are introduced. This design balances schema stability with representational flexibility, enabling long-term memory evolution without rigid schema constraints.

3) Similarity-Based Retrieval Strategy: We employ a pre-trained CLIP encoder [31] $E(\cdot,\cdot)$ to project the current task context—initial observation o_t and instruction l_t—into a joint embedding space. The retrieval is formalized as:

$$\mathcal{H}_t = \text{Top-}k\Big(\text{Sim}\big(E(o_t, l_t), \{E_i \mid e_i \in \mathcal{M}\}\big)\Big) \tag{3}$$

$\mathcal{H}_t$ encompasses both success exemplars and failure precedents, providing the **positive guidance** and **negative counterfactuals** necessary for attribution-driven reasoning.

B. The Brain: Attribution-Driven LLM Orchestrator

The **Brain** module serves as the deliberative core of SOMA, responsible for diagnosing execution failures and orchestrating interventions through the Model Context Protocol.

1) Semantic Tool Discovery and Intent Alignment: The Brain identifies required interventions by evaluating the capability gap between the current state o_t and the retrieved prototypes $\mathcal{H}_t$. Each tool T_i in the pool is defined by a semantic description $\mathcal{D}_i$. The LLM performs a zero-shot matching process to select a candidate toolset:

$$\{T_1, T_2, \ldots, T_k\} = \text{Match}\big(\text{Gap}(o_t, \mathcal{H}_t), \{\mathcal{D}_i\}_{i=1}^{N}\big) \tag{4}$$

where the selection is based on the alignment between the tool's functional description and the diagnosed cause of execution failure.

2) Knowledge-Augmented Parameterization: Following tool selection, The Brain derives execution parameters θ_i from the multi-modal context discrepancy Δ:

$$\theta_i = f_i(\Delta), \quad \Delta = \text{Extract-Discrepancy}(o_t, l_t, \mathcal{H}_t) \tag{5}$$

where f_i denotes the mapping function for tool-specific values. The final output is a set of parameterized interventions:

$$\mathcal{T}^* = \{(T_1, \theta_1), (T_2, \theta_2), \ldots, (T_k, \theta_k)\} \tag{6}$$

which grounds the intervention strategy in the quantitative differences between the current and reference trajectories.

3) Chain-of-Intervention Synthesis with Blank Inference: Before final sequencing, the Brain performs **Blank Inference** to validate the effectiveness of candidate tools. For each $(T_i, \theta_i) \in \mathcal{T}^*$, the Brain simulates the intervention's impact on the VLA's internal attention. An intervention is considered valid only if the post-simulation attention A'_t demonstrates improved semantic alignment and reduced spatial entropy:

$$\text{Valid}(T_i) = \begin{cases} \text{True}, & \Psi(A'_t, A_t, A_e, l_t) = \text{Succ} \\ & \wedge\ H(A'_t) < H(A_t), \quad \text{s.t. } \exists e \in \mathcal{H}_t \\ \text{False}, & \text{otherwise} \end{cases} \tag{7}$$

where Ψ denotes an LLM-based predicate that evaluates whether the simulated attention A'_t exhibits a superior focus on the target object compared to the initial A_t, using the attention map A_e of a retrieved experience $e \in \mathcal{H}_t$ as a topological reference. H represents spatial entropy. Tools that fail to yield more concentrated attention (i.e., $H(A'_t) \geq H(A_t)$ or $\Psi =$ Failure) are pruned from the intervention chain. The Brain then synthesizes only the validated tools into an ordered intervention chain:

$$\begin{aligned} \Phi_{\text{mcp}} &= \text{Orchestrate}(\mathcal{T}^*_{\text{valid}}) \\ &= \big\langle (T_1, \theta_1) \rightarrow (T_2, \theta_2) \rightarrow \cdots \rightarrow (T_k, \theta_k) \big\rangle. \end{aligned} \tag{8}$$

The orchestration follows two governing principles:

1) **Sensory-First Principle**: Perceptual corrections (e.g., *Paint-to-Action*, *Eraser*) are executed before physical recovery (e.g., *Encore*), preventing repeated failures due to misperception.
2) **Logical Dependency Principle**: Tools that eliminate causal interference are applied prior to subsequent reasoning or execution steps.

Example: For a co-diagnosed visual distribution shift and execution stagnation, the Brain generates the sequence:

$$\text{Paint-to-Action} \rightarrow \text{Encore}$$

This ordering ensures perceptual rectification precedes physical rollback. Finally, Φ_{mcp} is sent to the Limbs for serial execution, closing the perception-correction-recovery loop.

C. The Limbs: MCP Intervention for Agentic Execution

The **Limbs** module translates the intervention sequence Φ_{mcp} into perceptual or physical actions. Based on common VLA failure modes—such as distribution shifts and causal distractions—we define an initial suite of five specialized MCP tools. This modular architecture supports "plug-and-play" extensibility, allowing new tools to be integrated as novel failure archetypes are identified.

1) Paint-to-Action: Visual Domain Shift : Paint-to-Action performs non-parametric domain adaptation:

$$o'_t = \text{Paint-to-Action}(o_t, \theta_{\text{dist}}), \tag{9}$$

where θ_{dist} denotes the tool-specific execution parameter of Paint-to-Action generated in the Brain stage.

2) Eraser: Causal Confusion : Eraser removes irrelevant interference objects via tool-specific parameters θ_{conf}:

$$o'_t = \text{Eraser}(o_t, \theta_{\text{conf}}). \tag{10}$$

The Brain derives θ_{conf} by identifying task-interfering objects through a comparison of o_t with historical experiences $e \in \mathcal{H}_t$, thereby purifying the scene.

3) Prompt-Refiner: Linguistic Ambiguity: Prompt-Refiner normalizes semantic instructions to mitigate linguistic noise:

$$l'_t = \text{Prompt-Refiner}(l_t, \theta_{\text{lang}}), \tag{11}$$

where θ_{lang} is derived by distilling actionable intent from ambiguous commands l_t, using successful grounding templates in $\mathcal{H}_t$. It rewrites noisy inputs into standardized instructions l'_t that align with the VLA policy's training distribution.

4) Chaining-Step: Temporal Compounding : Chaining-Step implements horizon reduction:

$$A'_{seq} = \text{Chaining-Step}(A_{seq}, \mathcal{K}), \tag{12}$$

splitting long-horizon tasks into sub-tasks and resetting the VLA's context window.

5) Encore: Execution Stagnation: Encore resolves execution deadlocks through closed-loop key-state recovery:

$$s'_t = \text{Encore}(s_t, \theta_{\text{stuc}}), \tag{13}$$

where s_t is the current state and θ_{stuc} denotes recovery parameters derived from the Brain's diagnosis. By referencing failure precedents $e_{\text{fail}} \in \mathcal{H}_t$, Encore facilitates adaptive physical recovery without altering the VLA's underlying control logic.

D. Evolution: Dual-Stage Memory Consolidation

The **Evolution** module consolidates the latest execution trajectory into the dual-memory bank $\mathcal{M}$ using a two-stage attribution mechanism.

1) Stage 1: Online Reactive Attribution: Upon task failure, the Brain performs online reactive attribution:

$$e_{new} = \text{Online-Attribution}(o_{fail}, T_{raw}), \tag{14}$$

and stores the new experience:

$$\mathcal{M}_{fail} = \mathcal{M}_{fail} \cup \{e_{new}\}. \tag{15}$$

2) Stage 2: Offline Comparative Reflection: Offline comparative reflection retrieves the top-N similar historical failures $\mathcal{E}_{\text{sim}} = \{e^1_{\text{sim}}, \ldots, e^N_{\text{sim}}\}$ and conducts a joint cross-task diagnosis to facilitate mutual refinement. This process is formalized as a set-to-set transformation:

$$\{e^*_{\text{new}}, e^{1*}_{\text{sim}}, \ldots, e^{N*}_{\text{sim}}\} = \text{Offline-Reflection}(e_{\text{new}}, \mathcal{E}_{\text{sim}}), \tag{16}$$

where each e^* denotes an updated experience tuple. In this stage, the LLM performs a differential analysis across the batch, rectifying historical failure descriptions and optimizing correction plans in light of the new evidence. This ensures that the dual-memory bank $\mathcal{M}$ undergoes iterative self-correction, rather than simple monotonic expansion.

Fig. 4: **Performance comparison of SOMA across diverse robotic foundation models and OOD challenges.** The results demonstrate that SOMA consistently yields substantial performance gains for all base models across visual, linguistic, and long-horizon tasks.

IV. Experiment

In this section, we evaluate the effectiveness of the proposed SOMA framework by conducting experiments across various foundation models on a series of custom benchmarks. These tasks are specifically designed to represent Out-of-Distribution (OOD) scenarios. Additionally, we perform a targeted ablation study to validate the efficacy of our evolution mechanism. Our results demonstrate that SOMA effectively bridges the performance gap in diverse scenarios, providing measurable gains in the zero-shot capabilities and robustness of the underlying robotic policies. To ensure statistical significance—addressing the trial-count limitations often inherent in physical robot experiments—we prioritize **rigorous testing** within a high-fidelity simulation environment. By conducting approximately 200 trials for each task, we provide a more reliable assessment of the framework's adaptive reasoning, laying a solid foundation for future physical deployment.

A. Experimental Setup

Simulation Environment and Task Design. To assess zero-shot robustness under distributional shifts that standard benchmarks often overlook, we introduce **LIBERO-SOMA**. Rather than arbitrary modifications, this suite formalizes five fundamental OOD challenge dimensions commonly encountered in both simulation and real-world robotic tasks. By reconfiguring assets within the original LIBERO scenarios [32], we provide a controlled environment to evaluate an agent's ability to trigger adaptive interventions for recurring failure modes without task-specific fine-tuning.

- **Visual Focus**: Selecting a target from five identical, novel bowls. By substituting *LIBERO-Spatial* targets with indistinguishable assets, we evaluate **referential grounding** under fine-grained visual ambiguity.
- **Clutter Removal**: Grasping targets amidst dense distractors. We augment *LIBERO-Spatial* with high-entropy object clusters to test **operational focus** and interference mitigation in crowded scenes.
- **Noisy Prompt**: Intent extraction from informal commands (e.g., *"Umm... get that red squeezy thing"*). By injecting linguistic noise into *LIBERO-Object* instructions, we assess **linguistic robustness** and OOD semantic mapping.
- **Execution Fragility**: Critical maneuvers where minor slips trigger failure. We curate unstable sub-tasks from *LIBERO-Object* to evaluate **operational stability** against mechanical uncertainties.
- **Task Chaining**: Long-horizon sorting of multiple objects into a basket. By sequencing atomic *LIBERO-Object* actions, this task evaluates **long-term reasoning** and **error resilience** in sequential workflows.

In-Context Initialization and Metrics. To provide the agent with a repertoire of prior semantic experiences, we initialize the Dual-Memory RAG via a preliminary inference pass over the standard LIBERO-Object and Spatial sets. *Notably, this process acts as a non-parametric "cold start" that populates the memory bank with generic execution traces, which does not involve any gradient-based training or exposure to the specific OOD perturbations encountered in the evaluation phase.* This is followed by 199 randomized test rollouts per task, where object poses and task sequences are fully shuffled to prevent bias. Performance is quantified by **overall completion rates**, supplemented by **intermediate success rates** to provide a granular analysis of progress in long-horizon tasks.

Baselines. We evaluated the SOMA framework across a diverse range of foundation models, including π_0 [21], $\pi_{0.5}$ [23] and SmolVLA [33]. The results show general improvements in success rates across various OOD scenarios, indicating that SOMA contributes effectively to the generalization capabilities of different underlying robotic policies.

B. Main Results

We evaluate the **SOMA** framework across three foundation models—*pi05-LIBERO*, *smolvla-LIBERO*, and *pi0-LIBERO*—over five distinct challenge dimensions. As illustrated in Fig. 4, the SOMA-augmented policies show consistent performance improvements relative to the base models in all evaluated scenarios.

Environmental and Linguistic Variations. Base models exhibit sensitivity to sensory and linguistic interference. In *Visual Focus* and *Clutter Removal* tasks, the success rates of *pi05-LIBERO* are 1.7% and 31.8%, respectively. Following the integration of SOMA, these rates increase to 76.9% and 74.4%, suggesting that the framework aids in isolating task-relevant features from environmental noise. Similarly, *smolvla-LIBERO* achieves 98.3% under *Noisy Prompts* (up from 0.0%) via SOMA's linguistic normalization.

Execution Stability and Long-Horizon Tasks. A primary challenge identified for base models is **Execution Fragility**, where minor operational slips frequently lead to task failure. SOMA addresses this by providing recovery interventions, resulting in success rates exceeding 87% for all models in this category. For *Task Chaining*, while base models encounter compounding errors that reduce completion rates in subsequent stages, SOMA-augmented models demonstrate higher temporal consistency. Notably, *pi05-LIBERO* maintains a 96.0% success rate through to the final sub-task.

Framework Generalization. The observed improvements across diverse model architectures and scales indicate that SOMA is a model-agnostic framework. These results suggest that the proposed architecture can facilitate the adaptation of general-purpose foundation models to manipulation tasks requiring higher execution precision and robustness.

C. Ablation Study: Impact of the Evolution Mechanism

With the Eyes, Brain, and Limbs modules' effectiveness confirmed in Section IV-B, we further dissect the functional utility of the **Dual** nature within our Dual-Memory RAG system. As formulated in Section III-A, this analysis is two-fold: we first isolate the Dual-Memory Bank's impact on execution efficiency, then evaluate how the Two-Stage Attribution—the core process of the **Evolution** mechanism—contributes to reasoning depth.

1) Efficacy of the Dual-Memory Bank: To verify the necessity of storing both successful and failed experiences for anchoring LLM decisions, we conduct an ablation using the *Turns to Success* metric. This metric quantifies the number of reasoning-intervention loops required to achieve an optimal intervention plan. As illustrated in Fig. 5, we compare four memory configurations: (i) **No Memory**, (ii) **Failure Only**, (iii) **Success Only**, and (iv) **SOMA (Dual)**.

As shown in Fig. 5, the baseline without memory consistently hits the maximum execution timeout ($Mean = 7.40$), proving that zero-shot reasoning is insufficient for OOD recovery. While *Success-Only* memory reduces average turns to 1.47, it retains significant stochastic variance. In contrast, **SOMA (Dual)** converges to a near-optimal **1.07** turns. This suggests that the dual-memory bank functions as a "visionary reference," transforming the agent from a trial-and-error explorer into a deterministic orchestrator by explicitly mapping both reachable paths and forbidden zones.

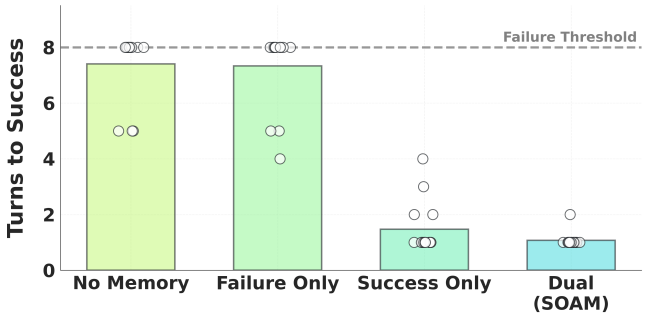


Fig. 5: **Ablation Study:** Distribution of reasoning turns required to reach the target intervention score.

2) Impact of Two-Stage Attribution (Evolution): While the dual-memory bank provides the raw data, the quality of these experiences is governed by the **Evolution** module's iterative attribution. We evaluate this processvia three configurations: **No RAG** (baseline), **Limited RAG** (single-stage online attribution), and **Rich RAG** (SOMA's dual-stage consolidation).

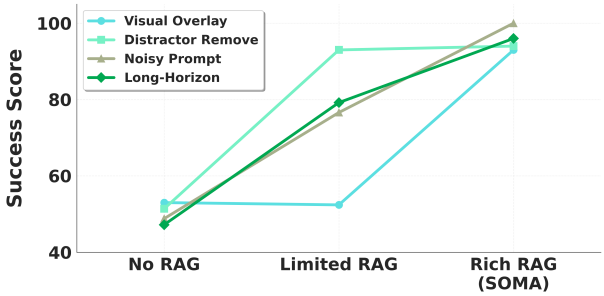


Fig. 6: **Ablation Study:** Evolution of qualitative reasoning depth scores across configurations.

Quantitative Performance and Stability We evaluate our framework on a variant of the "Clutter Removal" task using Pi-0.5 as the base policy. As shown in Table I, success rates scale with the richness of RAG information. The No RAG baseline yields only 19.20%; notably, comparison with Section IV-B suggests that relying solely on LLM guidance without RAG can even be counterproductive, potentially introducing negative biases. Performance improves to 48.80% with Limited RAG, though its tool chain reveals a reliance on redundant steps like unnecessary background replacement. In contrast, Rich RAG (SOMA) achieves a peak success rate of 60.10% through a streamlined reasoning chain, demonstrating superior strategic efficiency and precision.

Qualitative Evaluation of Reasoning Depth To assess the model's cognitive depth, we use Qwen3-VL-32B to score the logical coherence of generated instructions. As illustrated in Fig. 6, Rich RAG consistently secures the highest ratings across all dimensions. This advantage stems from optimized reasoning trajectories that prioritize critical intervention points, thereby reducing ineffective action se-

TABLE I: **Ablation Study:** Comparison of reasoning logic and success rates across RAG configurations.

RAG Config	Reasoning	MCP Tool	Action Result	Task Image	SR (%)
No-RAG	Ambiguity in target ID due to identical textures.	$o'_t = \text{Paint-to-Action}(o_t, \theta_{\text{dist}})$	Highlight bowl		19.2
	Potential detection errors from plate shadows.	`none`	—		
	Clean background, no distractors found.	`none`	—		
Limited-RAG	Edge proximity causes occlusion; highlight target.	$o'_t = \text{Paint-to-Action}(o_t, \theta_{\text{dist}})$	Highlight bowl		48.8
	1. **Past failures** suggest high mis-ID risk. 2. Focus by **removing** identical distractors.	$o''_t = \text{Eraser}(o'_t, \theta_{\text{conf}})$	Remove 4 dist.		
	Background replace to reduce visual noise.	$o'''_t = \text{Eraser}(o''_t, \theta_{\text{conf}})$	BG Replacement		
Rich-RAG	1. High risk from **distractors via failure memory**. 2. **Memory** dictates explicit removal of each bowl. 3. Strategic preservation of target and plate. 4. Use precise position to guide policy.	$o'_t = \text{Eraser}(o_t, \theta_{\text{conf}})$	Remove 4 dist.		60.1
	Distraction-free; no visual overlay needed.	`none`	–		

quences. These results indicate that high-quality RAG data directly strengthens the model's grasp of task causality, forming a more robust tool-invocation chain that aligns with the observed quantitative gains.

V. CONCLUSIONS

This work investigates the performance constraints of zero-shot VLA generalization, observing that failures in out-of-distribution (OOD) scenarios often arise from systemic coordination and perceptual challenges rather than inherent motor limitations. To address this, we propose SOMA, a strategic orchestration and memory-augmented agentic framework that upgrades frozen VLA policies without parameter fine-tuning. By integrating the Model Context Protocol (MCP) for dynamic tool orchestration and a recursive cognitive cycle of retrieval, reasoning, and consolidation, SOMA enables a self-evolving capability for experience reuse and failure-aware intervention. Extensive evaluations across various VLA backbones demonstrate that SOMA significantly enhances zero-shot success rates and robustness under diverse OOD challenges. Future work will explore scaling the repository of MCP-based tools to encompass a more diverse spectrum of OOD challenges. Additionally, we aim to leverage frontier multimodal models with superior cross-modal reasoning to refine the granularity of failure attribution, further enhancing the framework's adaptive capabilities.

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